Pardis Taghavi

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Website Google Scholar GitHub LinkedIn

Research Focus

Final-year PhD student at Texas A&M University working on efficient and controllable video generation and world models. Research spans sparse attention, action conditioning, and foundation model distillation, with broader interests in physical AI.

Education

Texas A&M University January 2022 – Present
PhD in Mechanical Engineering
GPA 4.0/4.0 · Advisors Gaurav Pandey and Reza Langari

Politecnico di Torino August 2018 – August 2021
BS in Mechanical Engineering
110/110, *summa cum laude*

Research Experience

Graduate Researcher **January 2022 – Present**
Texas A&M University *College Station, TX*

Computer Vision & Robotics Lab / TACO Group March 2025 – Present
Research with Gaurav Pandey and Zhengzhong Tu

- Developed training-free sparse attention with residual correction for video generation and world models, achieving 1.48–2.61× end-to-end inference speedups across four models from the Cosmos, Wan, and HunyuanVideo families while preserving generation quality.
- Developed learned state correction for interactive world models, enabling action changes during video generation without restarting sampling.
- Investigate long-horizon video generation and controllable world models, focusing on temporal consistency and responsiveness to changing action inputs.

Visual Intelligence Lab June 2024 – March 2025
Research with Shu Kong

- Developed a semi-supervised foundation model distillation framework for training compact visual recognition models from limited labeled data.

Autonomous Driving Research January 2022 – May 2024
Research with Reza Langari

- Extensively fine-tuned and evaluated visual perception models for autonomous driving across object detection, semantic segmentation, instance segmentation and depth estimation tasks.
- Led the autonomous driving perception research team, developing a modular pipeline integrating cameras, LiDAR, and IMU data for scene understanding in rural and urban environments, supporting over 40 miles of autonomous driving with our autonomous vehicle.

Industry Experience

AI Engineer Intern **May 2025 – September 2025**
Rivian *Palo Alto, CA*

- Developed and evaluated a GRPO-based reinforcement learning framework for model post-training, improving robustness under noisy in-vehicle conditions compared with supervised fine-tuning.
- Designed a curriculum learning schedule to stabilize RL training and improve generalization.

Selected Publications

[1] **Partition the Support, Reconstruct the Residual: Training-Free Sparse Attention for Video Generation and World Models**
Pardis Taghavi, Reza Langari, Gaurav Pandey
arXiv, 2026 · Under review · Paper · Project

[2] **ActionSplice: In-Flight Action Editing for Interactive World Models**
Pardis Taghavi, Tingyu Guo, Jonas Lossner, Gaurav Pandey, Reza Langari
arXiv, 2026 · Under review · Paper · Project

[3] **The Pulse of Motion: Measuring Physical Frame Rate from Visual Dynamics**
Xiangbo Gao, Mingyang Wu, Siyuan Yang, Jiongze Yu, **Pardis Taghavi**, Fangzhou Lin, Zhengzhong Tu
WACV, 2027 · Paper · Project

[4] **NaviDriveVLM: Decoupling High-Level Reasoning and Motion Planning for Autonomous Driving**
Ximeng Tao, **Pardis Taghavi**, Dimitar Filev, Reza Langari, Gaurav Pandey
arXiv, 2026 · Under review · Paper · Code

[5] **Training a Student Expert via Semi-Supervised Foundation Model Distillation**
Pardis Taghavi, Tian Liu, Renjie Li, Reza Langari, Zhengzhong Tu
CVPRW, 2025 · Paper

[6] **SwinMTL: A Shared Architecture for Simultaneous Depth Estimation and Semantic Segmentation from Monocular Camera Images**
Pardis Taghavi, Reza Langari, Gaurav Pandey
IROS, 2024 · Paper · Code

[7] **Spec Kit Agents: Context-Grounded Agentic Workflows**
Pardis Taghavi, Santosh Bhavani
arXiv, 2026 · Paper

Technical Skills

Generative modeling	Video diffusion models, diffusion transformers (DiT), autoregressive world models, flow matching, action conditioning
Model training	Fine-tuning, reinforcement learning (GRPO), knowledge distillation, semi-supervised learning
ML frameworks & libraries	PyTorch, Hugging Face Diffusers, vLLM, FlashAttention, FlashInfer, NVIDIA Cosmos Framework
Programming	Python, C/C++
Systems & simulation	Linux, CUDA, Slurm, ROS, CARLA

Honors and Awards

NSF Research Subaward 2024
Remote testing and validation of autonomous vehicle perception at Mcity

TOPolito Scholarship 2018 – 2021
Awarded to the top 10 students of the engineering school

Academic Service

Committee · CVPR Workshop on Multi-Agent Embodied Intelligent Systems, 2026

Reviewer · CVPR, ECCV, NeurIPS